

JEE MAINS QFT-19 2025



JEE, let's talk about JEE.

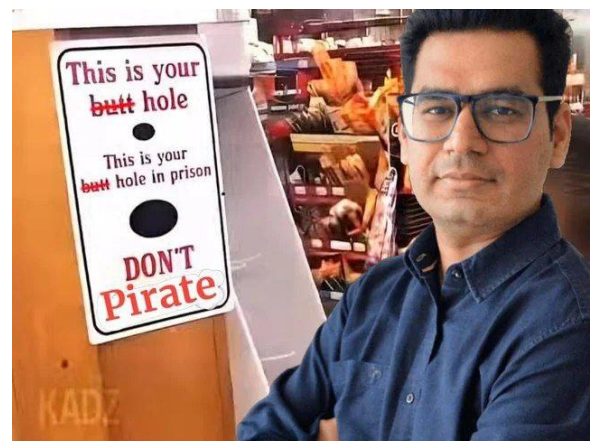
Is it anything and everything you have hoped for?

Or do the failure haunt you....

I know the failure haunts you.... ✨



By: C I P H E R



Question Paper

Physics Single Correct (Maximum Marks: 80)

Question No. 1 (Only One Option Correct Type)

The free space inside a current-carrying toroid is filled with a material of susceptibility 2×10^{-2} . The percentage increase in the value of the magnetic field inside the toroid will be

- (a) 2%
- (b) 0.2%
- (c) 0.1%
- (d) 1%

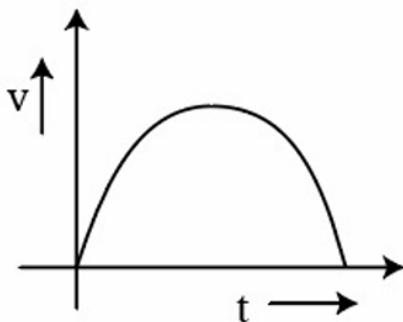
Question No. 2 (Only One Option Correct Type)

The size of the image of an object, which is at infinity, as formed by a convex lens of focal length 30 cm is 2 cm. If a concave lens of focal length 20 cm is placed between the convex lens and the image at a distance of 26 cm from the lens, the new size of the image is

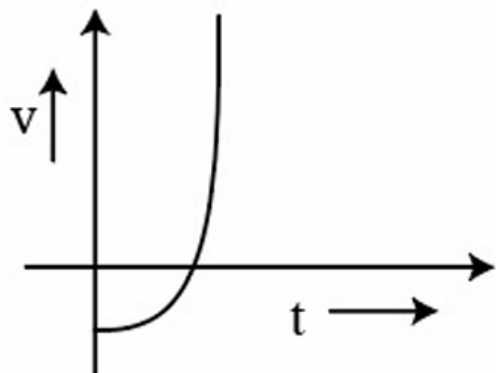
- (a) 1.25 cm
- (b) 2.5 cm
- (c) 1.05 cm
- (d) 2 cm

Question No. 3 (Only One Option Correct Type)

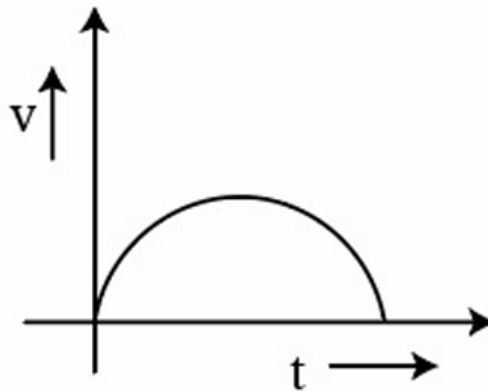
The displacement-time ($x - t$) graph of a body is given as



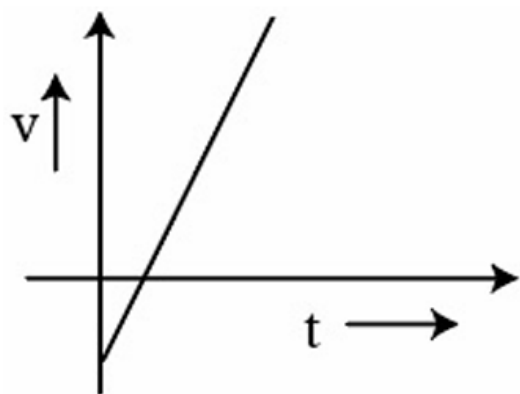
The corresponding velocity-time ($v - t$) graph will be



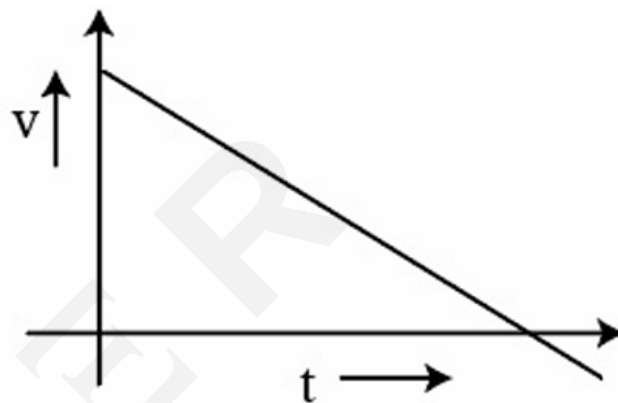
(A)



(B)



(C)



(D)

Question No. 4 (Only One Option Correct Type)

A body of mass 1 kg begins to move from rest under the action of a time-dependent force

$$\vec{F} = (2t\hat{i} + 3t^2\hat{j}) \text{ N},$$

where $\hat{i}$ and $\hat{j}$ are unit vectors along the x and y axes. What power will be developed by the force at time t ?

- (a) $(2t^2 + 3t^3) \text{ W}$
- (b) $(2t^2 + 4t^4) \text{ W}$
- (c) $(2t^3 + 3t^4) \text{ W}$
- (d) $(2t^3 + 3t^5) \text{ W}$

Question No. 5 (Only One Option Correct Type)

Names of units of some physical quantities are given in List-I, and their corresponding dimensional formulae are given in List-II. Match the correct pairs.

List-I (Physical Quantities)

List-II (Dimensional Formulae)

(A) Surface Tension

(i) $[ML^2T^{-1}]$

(B) Planck's Constant

(ii) $[ML^0T^{-2}]$

(C) Boltzmann Constant

(iii) $[ML^2T^{-2}K^{-1}]$

(D) Magnetic Flux

(iv) $[ML^2T^{-2}A^{-1}]$

- **(a)** (ii) (i) (iii) (iv)
- **(b)** (i) (iii) (ii) (iv)
- **(c)** (iv) (i) (iii) (ii)
- **(d)** (ii) (iii) (i) (iv)

Question No. 6 (Only One Option Correct Type)

The number of turns, cross-sectional area and length for four solenoids are given in the following table.

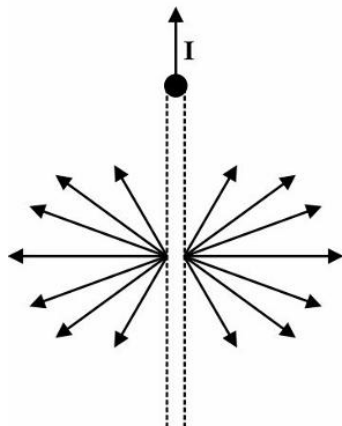
Solenoid	Total Turns	Area	Length
1	2 N	2A	ℓ
2	2 N	A	ℓ
3	3 N	3A	2ℓ
4	2 N	2A	$\ell/2$

The solenoid with maximum self inductance is :

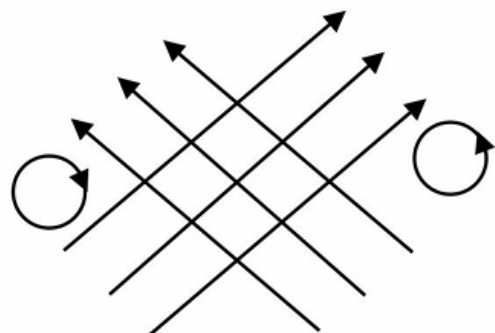
- **(a)** 1
- **(b)** 2
- **(c)** 3
- **(d)** 4

Question No. 7 (Only One Option Correct Type)

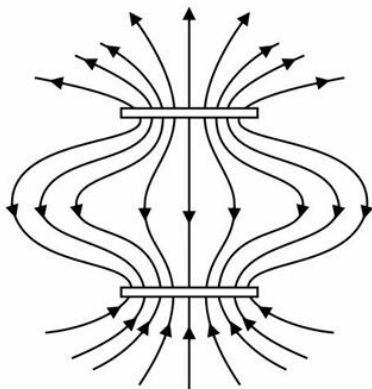
Point out the correct direction of the magnetic field in the given figures.



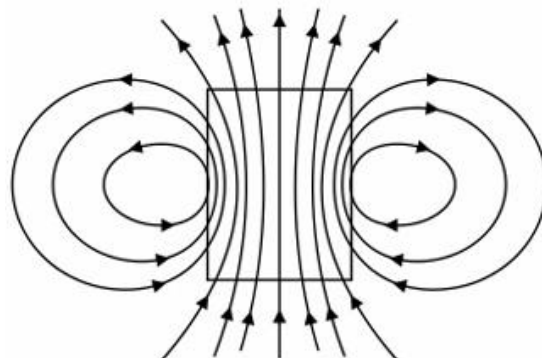
(A)



(B)



(C)



(D)

Question No. 8 (Only One Option Correct Type)

Two persons A and B wear glasses of optical powers (in air) $P_1 = +2D$ and $P_2 = +1D$ respectively. The glasses have a refractive index of 1.5. Now they jump into a swimming pool and look at each other. B appears to be present at a distance of 2 m (from A) to A . A appears to be present at a distance of 1 m (from B) to B . Find out the refractive index of water in the swimming pool, in the form $\frac{X}{10}$, and fill the value of X .

- (a) 15
- (b) 18
- (c) 12
- (d) 20

Question No. 9 (Only One Option Correct Type)

A balloon of volume 1500 m^3 and weighing 1650 kg with all its equipment is filled with He (density 0.2 kg m^{-3}). If the density of air be 1.3 kg m^{-3} , the pull on the rope tied to the balloon will be

- (a) 300 kg
- (b) 1950 kg
- (c) 1650 kg
- (d) zero

Question No. 10 (Only One Option Correct Type)

Assertion: In S.H.M., the velocity and displacement of the particle are in the same phase.

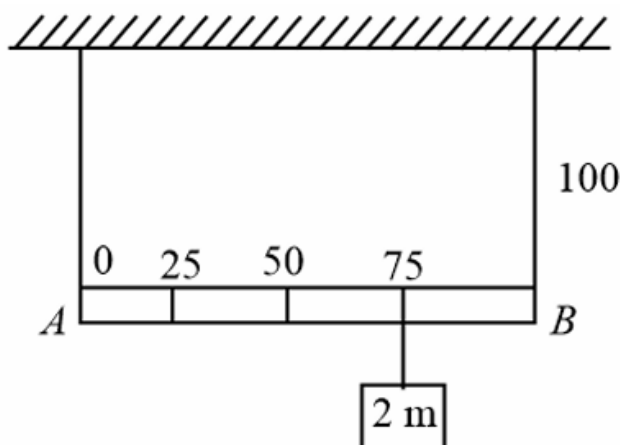
Reason: Velocity is the ratio of displacement to the time taken.

- (a) Assertion is True, Reason is True; Reason is a correct explanation for Assertion
- (b) Assertion is True, Reason is True; Reason is not a correct explanation for Assertion

- **(c)** Assertion is True, Reason is False
- **(d)** Assertion is False, Reason is True

Question No. 11 (**Only One Option Correct Type**)

Shown in the figure is a rigid and uniform one-meter-long rod AB held in a horizontal position by two strings tied to its ends and attached to the ceiling. The rod is of mass m and has another weight of mass $2m$ hung at a distance of 75 cm from A . The tension in the string at A is:



- **(a)** $0.5\ mg$
- **(b)** $2\ mg$
- **(c)** $0.75\ mg$
- **(d)** $1\ mg$

Question No. 12 (**Only One Option Correct Type**)

A simple pendulum has a time period T_1 when on the Earth's surface and T_2 when taken to a height R above the Earth's surface, where R is the radius of the Earth. The value of $\frac{T_2}{T_1}$ is

- **(a)** 1
- **(b)** $\sqrt{2}$
- **(c)** 4
- **(d)** 2

Question No. 13 (**Only One Option Correct Type**)

If the magnetic field in a plane electromagnetic wave is given by

$$\vec{B} = 3 \times 10^{-8} \sin (1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{y} \text{ T},$$

then the expression for the electric field is:

- (a) $\vec{E} = 60 \sin (1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{k} \text{ V/m}$
- (b) $\vec{E} = 9 \sin (1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{k} \text{ V/m}$
- (c) $\vec{E} = 3 \times 10^{-8} \sin (1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{y} \text{ V/m}$
- (d) $\vec{E} = 3 \times 10^{-8} \sin (1.6 \times 10^3 x + 48 \times 10^{10} t) \hat{z} \text{ V/m}$

Question No. 14 (Only One Option Correct Type)

Which of the following is correct for the molecules of a gas in thermal equilibrium?

- (a) All have the same speed
- (b) All have different speeds which remain constant
- (c) They have a certain constant average speed
- (d) They do not collide with one another

Question No. 15 (Only One Option Correct Type)

To observe the diffraction phenomenon, the minimum size of the hole is needed for which of the following waves?

- (a) Microwaves
- (b) Radio waves
- (c) Sound waves
- (d) Visible light

Question No. 16 (Only One Option Correct Type)

An open carriage in a goods train is moving with a uniform velocity of 10 m/s. If the rain adds water with zero velocity at the rate of 5 kg/s, then the additional force applied by the engine to maintain the same velocity of the train is

- (a) 0.5 N
- (b) 2.0 N
- (c) 50 N
- (d) 25 N

Question No. 17 (Only One Option Correct Type)

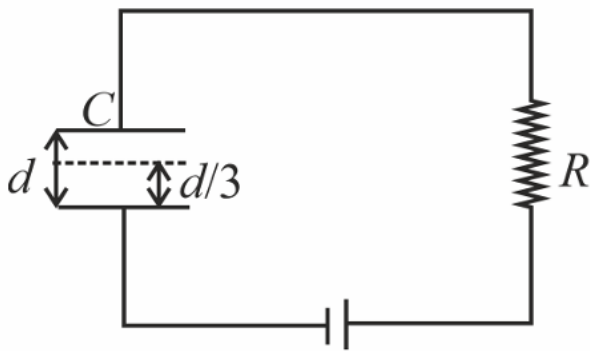
The temperature at which a molecule of nitrogen will have the same rms velocity as a molecule of oxygen at 127°C is (Mark answer in Kelvin) (Round off answer to the nearest integer).

- (a) 350 K

- (b) 250 K
- (c) 330 K
- (d) 230 K

Question No. 18 (Only One Option Correct Type)

A parallel plate capacitor C with plates of unit area and separation d is filled with a liquid of dielectric constant $K = 2$, the level of the liquid is $\frac{d}{3}$, initially. Suppose the liquid level decreases at a constant speed v , the time constant as a function of time is



- (A) $\frac{6\varepsilon_0 R}{5d + 3vt}$
- (B) $\frac{(15d + 9vt)\varepsilon_0 R}{2d^2 - 3dvt - 9v^2 t^2}$
- (C) $\frac{6\varepsilon_0 R}{5d - 3vt}$
- (D) $\frac{(15d - 9vt)\varepsilon_0 R}{2d^2 + 3dvt - 9v^2 t^2}$

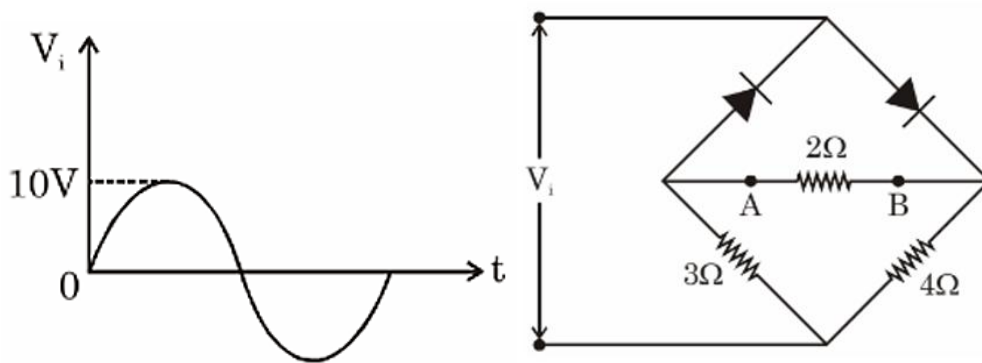
Question No. 19 (Only One Option Correct Type)

In a hypothetical Bohr hydrogen atom, if the mass of the electron is doubled, then the energy of the electron in the first orbit is

- (a) -27.2eV
- (b) -13.6eV
- (c) -6.8eV
- (d) -3.4eV

Question No. 20 (Only One Option Correct Type)

In the circuit shown in the figure, find the magnitude of the maximum potential difference (in V) between A and B .



- (A) 1.5
- (B) 2
- (C) 4
- (D) 0.5

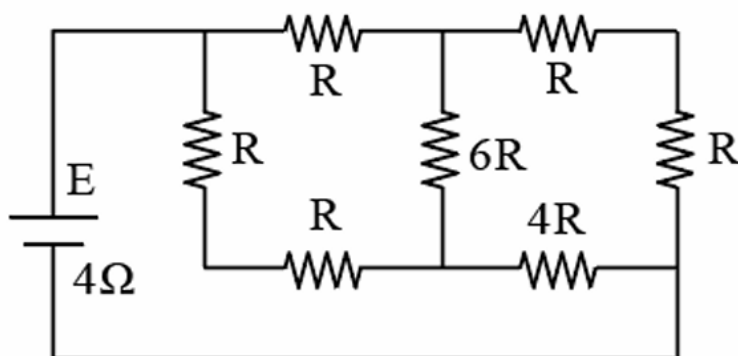
Physics Numerical (Maximum Marks: 20)

Question No. 1 (Numerical Type)

When a loop of wire is dipped in a wetting liquid and taken out, a liquid film is formed, and a loop of light inextensible thread is gently put on the liquid film. A hole is pricked inside the loop of thread. Due to the property of surface tension, the free surface of the liquid tries to minimize its surface area, and hence the area of the hole will be maximized. In the above description, the loop wire is a square of side 4 units. If the length of the thread is 15 units, the final surface area for the soap film on one side of the wire frame is $\frac{1}{K} \left(\frac{1}{4-\pi} \right)$. Find K .
(Neglect friction everywhere.)

Question No. 2 (Numerical Type)

A battery of internal resistance $4\ \Omega$ is connected to the network of resistances as shown. In order that the maximum power can be delivered to the network, the value of R in Ω should be



Question No. 3 (Numerical Type)

Two electric dipoles of moments p and $27p$ are placed in opposite directions on a line at a distance of 24 cm. The electric field will be zero at a point between the dipoles whose distance from the dipole of moment p is (mark answer in cm).

Question No. 4 (Numerical Type)

If a coil of 40 turns and area 4.0 cm^2 is suddenly removed from a magnetic field, it is observed that a charge of $2.0 \times 10^{-4} \text{ C}$ flows into the coil. If the resistance of the coil is 80Ω , the magnetic flux density is _____ T.

Question No. 5 (Numerical Type)

A liquid of bulk modulus k is compressed by applying an external pressure such that its density increases by 0.01%. The pressure applied on the liquid is $\frac{k}{N}$. Find the integral value of N .

Chemistry Single Correct (Maximum Marks: 80)

Question No. 1 (Only One Option Correct Type)

The volume of ethyl alcohol (density 1.15 g/cc) that has to be added to prepare 100 cc of 0.5 M ethyl alcohol solution in water is

- (a) 1.15 cc
- (b) 2 cc
- (c) 2.15 cc
- (d) 2.30 cc

Question No. 2 (Only One Option Correct Type)

Which of the following statements are correct?

- (i) Inductive effect and resonance effect are possible in chlorobenzene.
- (ii) Resonance effect dominates over inductive effect in anisole.

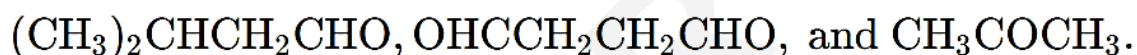
(iii) *p*-nitrobenzoic acid is less acidic than *m*-nitrobenzoic acid.

(iv) Diphenylamine is more basic than aniline.

- (a) (i), (ii), (iii), and (iv)
- (b) (i) and (ii)
- (c) (iii) and (iv)
- (d) (i) and (iii)

Question No. 3 (Only One Option Correct Type)

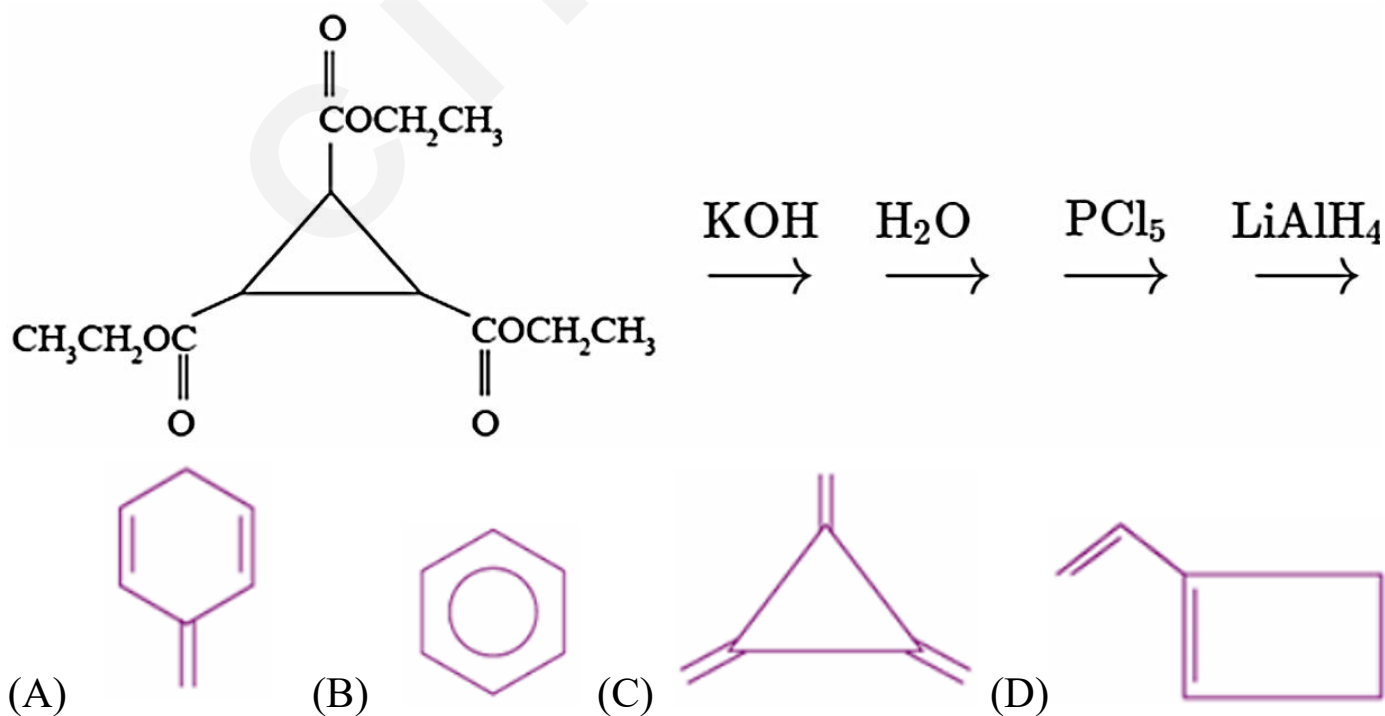
A hydrocarbon (A) C_nH_{2n-4} on ozonolysis gives



The value of n is

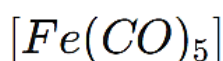
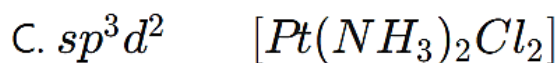
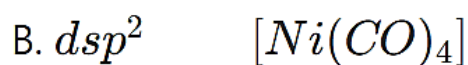
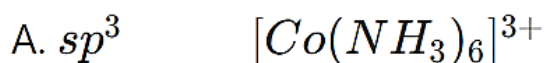
- (a) 12
- (b) 14
- (c) 16
- (d) 18

Question No. 4 (Only One Option Correct Type)



Question No. 5 (Only One Option Correct Type)

Column I	Column II
A. sp^3	$[Co(NH_3)_6]^{3+}$
B. dsp^2	$[Ni(CO)_4]$
C. sp^3d^2	$[Pt(NH_3)_2Cl_2]$
D. dsp^3	$[CoF_6]^{3-}$
	$[Fe(CO)_5]$

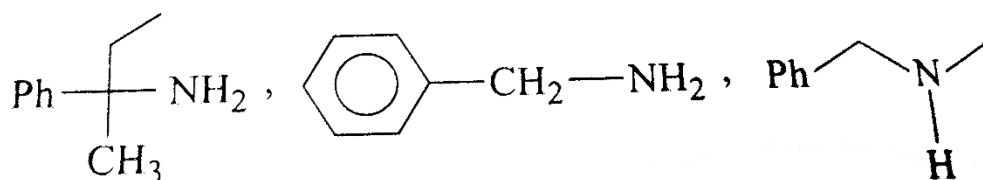
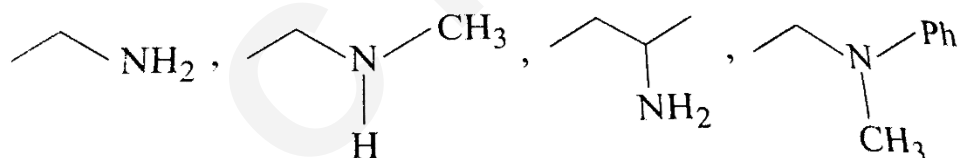


Match the following.

- (a) (v) (ii) (iv) (iii)
- (b) (ii) (iii) (iv) (v)
- (c) (ii) (iii) (i) (v)
- (d) (iii) (ii) (iv) (i)

Question No. 6 (Only One Option Correct Type)

Of the following amines, how many can be separated by Hoffmann's mustard oil reaction?



- (a) 3
- (b) 4
- (c) 5
- (d) 6

Question No. 7 (Only One Option Correct Type)

The half-life period of the radioactive element A is 10 days. The amount of A left at the end of the 11th day, starting with 1 mole of A, is $\left(\frac{1}{2}\right)^{\frac{n+6}{10}}$ mole. What is the value of n ?

- (a) 4
- (b) 6
- (c) 5
- (d) 8

Question No. 8 (Only One Option Correct Type)

The set of amphoteric oxides among ZnO, Ti_2O_3 , In_2O_3 , B_2O_3 , PbO, SnO_2 is

- (a) ZnO, Ti_2O_3 , PbO
- (b) ZnO, PbO, SnO_2
- (c) ZnO, In_2O_3 , SnO_2
- (d) ZnO, In_2O_3 , PbO

Question No. 9 (Only One Option Correct Type)

The solubilities of AgBr in water, in 0.01 M CaBr_2 , 0.01 M KBr and in 0.05 M AgNO_3 are S_1, S_2, S_3 and S_4 respectively. Then the order of solubilities is

- (a) $S_1 > S_2 > S_3 > S_4$
- (b) $S_1 > S_3 > S_2 > S_4$
- (c) $S_2 > S_1 > S_3 > S_4$
- (d) $S_4 > S_3 > S_1 > S_2$

Question No. 10 (Only One Option Correct Type)

Among the compounds Benzene, Carbon tetrachloride, Naphthalene, Benzoic acid, Isooctane, and Anthracene, how many can be purified by sublimation?

- (a) 3
- (b) 4
- (c) 2
- (d) 1

Question No. 11 (Only One Option Correct Type)

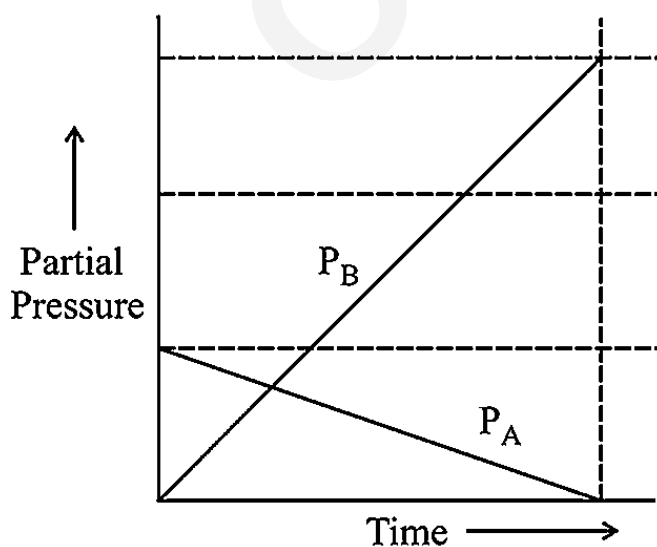
Consider the following sets of quantum numbers:

	n	l	m	s
(i)	3	0	0	$+\frac{1}{2}$
(ii)	2	2	1	$+\frac{1}{2}$
(iii)	4	3	-2	$-\frac{1}{2}$
(iv)	1	0	-1	$-\frac{1}{2}$
(v)	3	2	3	$+\frac{1}{2}$

- **(a)** (i), (ii), (iii), and (iv)
- **(b)** (ii), (iv), and (v)
- **(c)** (i) and (ii)
- **(d)** (ii), (iii), and (iv)

Question No. 12 (Only One Option Correct Type)

If for a reaction in which $A(g)$ converts to $B(g)$, the reaction carried out at constant V and T , results into the following graph, the order of reaction is:



- (A) 0
- (B) 1
- (C) 2

(D) 3

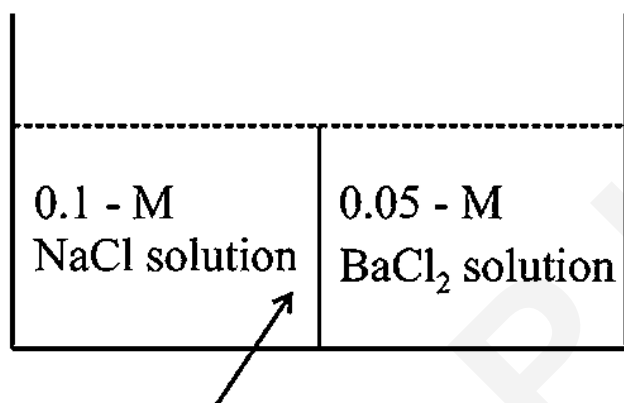
Question No. 13 (Only One Option Correct Type)

7.8 g of a compound having molecular formula C_6H_6 , on reacting with $CH_3COCl/AlCl_3$, gives 8.4 g of a product which has molecular formula C_8H_8O . Calculate the percentage yield of the product C_8H_8O . (Given atomic weights of H, C, and O are 1, 12, and 16 respectively.) (Mark answer to the nearest integer.)

- (a) 80
- (b) 65
- (c) 75
- (d) 70

Question No. 14 (Only One Option Correct Type)

Study the following figure and choose the correct options:



Semipermeable membrane

- (a) There will be no movement of any solution across the membrane.
- (b) $BaCl_2$ will flow towards the NaCl solution.
- (c) The osmotic pressure of 0.1 M NaCl is higher than the osmotic pressure of 0.05 M $BaCl_2$, assuming complete dissociation of the electrolyte.
- (d) NaCl will flow towards the $BaCl_2$ solution.

Question No. 15 (Only One Option Correct Type)

Statement I: Lothar Meyer arranged elements in increasing order of atomic number and observed periodic trends in their physical properties.

Statement II: Newlands' Law of Octaves states that every eighth element has properties similar to the first when elements are arranged in increasing order of atomic weight.

In the light of the above statements, choose the correct answer from the options given below:

- (a) Statement I is false, but Statement II is true
- (b) Both Statement I and Statement II are true

- **(c)** Statement I is true, but Statement II is false
- **(d)** Both Statement I and Statement II are false

Question No. 16 (**Only One Option Correct Type**)

Enzymatic reactions are given in Column I and enzymes in Column II.

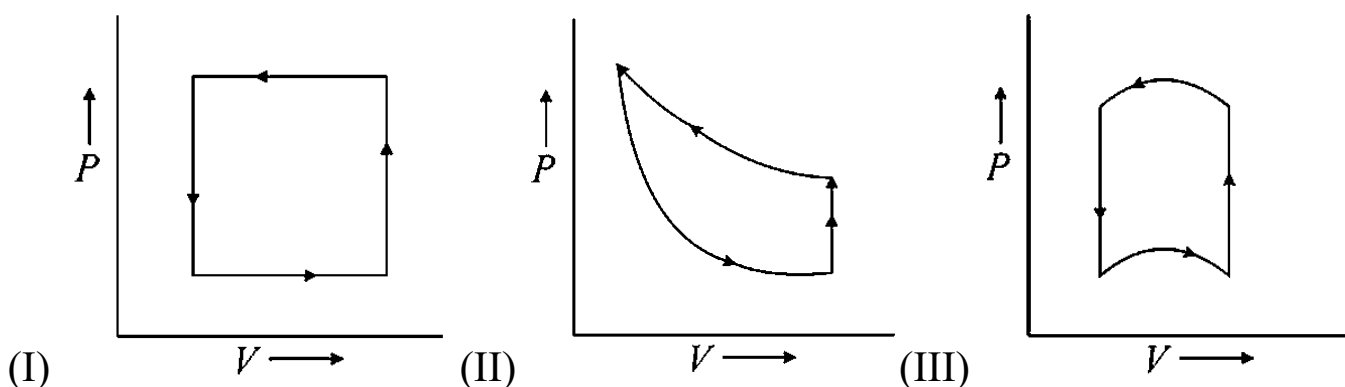
Column I	Column II
<i>A</i> Maltose $\rightarrow$ Glucose	<i>i</i> Zymase
<i>B</i> Sucrose $\rightarrow$ Glucose + Fructose	<i>ii</i> Pepsin
<i>C</i> Glucose $\rightarrow$ Ethyl alcohol + CO_2	<i>iii</i> Maltase
<i>D</i> Starch $\rightarrow$ Maltose	<i>iv</i> Invertase
<i>E</i> Proteins $\rightarrow$ Amino acids	<i>v</i> Diastase

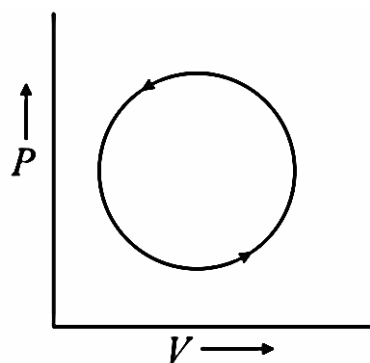
Choose the correct matching of enzymatic reaction and enzyme that catalyzes the correct reaction from the codes given below.

- **(a)** (*A*) – (*ii*), (*B*) – (*iv*), (*C*) – (*v*), (*D*) – (*iii*), (*E*) – (*i*)
- **(b)** (*A*) – (*iii*), (*B*) – (*iv*), (*C*) – (*i*), (*D*) – (*v*), (*E*) – (*ii*)
- **(c)** (*A*) – (*v*), (*B*) – (*iv*), (*C*) – (*ii*), (*D*) – (*i*), (*E*) – (*iii*)
- **(d)** (*A*) – (*v*), (*B*) – (*iii*), (*C*) – (*iv*), (*D*) – (*ii*), (*E*) – (*i*)

Question No. 17 (**Only One Option Correct Type**)

What will be the nature of change in internal energy in the case of processes shown below?





(IV)

- **(a)** +ve in all cases
- **(b)** -ve in all cases
- **(c)** -ve in I and III, and +ve in II and IV
- **(d)** Zero in all cases

Question No. 18 (Only One Option Correct Type)

When BaCl_2 is added to an aqueous salt solution, a white precipitate is obtained. How many anions among CO_3^{2-} , SO_3^{2-} , and SO_4^{2-} can be present in the solution?

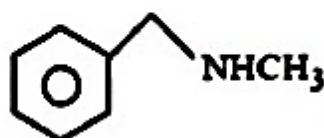
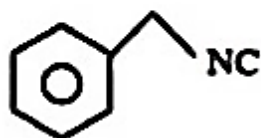
- **(a)** CO_3^{2-} and SO_3^{2-} only
- **(b)** CO_3^{2-} and SO_4^{2-} only
- **(c)** SO_4^{2-} and SO_3^{2-} only
- **(d)** CO_3^{2-} , SO_3^{2-} , and SO_4^{2-}

Question No. 19 (Only One Option Correct Type)

An organic compound X ($\text{C}_7\text{H}_7\text{Cl}$) when reacted with $\text{KCN}/\text{C}_2\text{H}_5\text{OH}$ gave a major product Y . Z is formed when Y is reduced with LiAlH_4 . What are Y and Z ?

Y

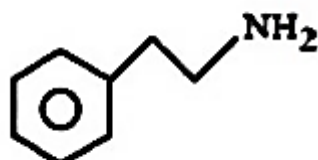
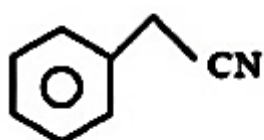
Z



(A)

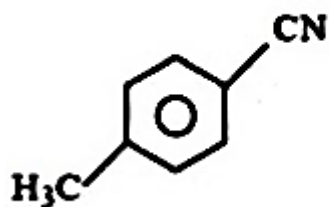
Y

Z

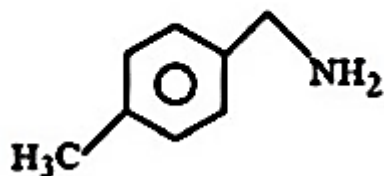


(B)

Y

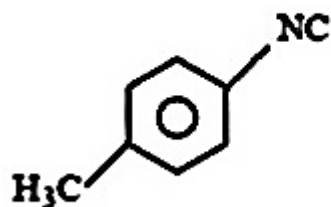


Z

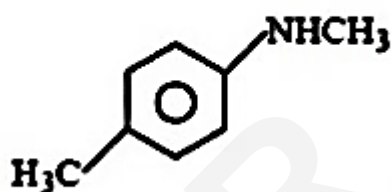


(C)

Y



Z



(D)

Question No. 20 (Only One Option Correct Type)

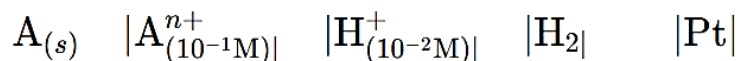
Pick out the incorrect statement(s) from the following.

1. Glucose exists in two different crystalline forms, α -D-glucose and β -D-glucose.
 2. α -D-glucose and β -D-glucose are anomers.
 3. α -D-glucose and β -D-glucose are enantiomers.
 4. Cellulose is a straight-chain polysaccharide made of only β -D-glucose units.
 5. Starch is a mixture of amylose and amylopectin, both containing unbranched chains of α -D-glucose units.
- (a) 1 and 2 only
 - (b) 2 and 3 only
 - (c) 3 and 4 only
 - (d) 3 and 5 only

Chemistry Numerical (Maximum Marks: 20)

Question No. 1 (Numerical Type)

Consider the following at 25°C:

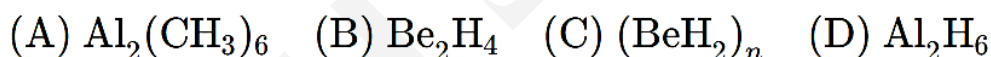


The e.m.f. of the cell is 0.89 V, and $E_{\text{A}|\text{A}^{n+}}$ is +0.98 V. The value of n is

$$\left[\text{Assume } \frac{2.303RT}{F} = 0.06 \right].$$

Question No. 2 (Numerical Type)

How many of the following compounds are associated with $sp^3 - s - sp^3$ type of overlap?



Question No. 3 (Numerical Type)

According to molecular orbital theory, the total number of bonding electron pairs in O_2 is

Question No. 4 (Numerical Type)

Silanes are silicon hydrides of the general formula $\text{Si}_n\text{H}_{2n+2}$ and have several applications. From the data given below, the bond dissociation enthalpy of the Si-Si bond (in kJ/mol) is

Given:

ΔH of the reaction $2\text{Si}(s) + 3\text{H}_2(g) \rightarrow \text{Si}_2\text{H}_6(g)$ is 80.3 kJmol^{-1}

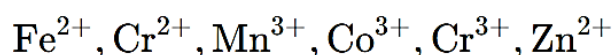
Bond dissociation enthalpy for $\text{H} - \text{H} = 436 \text{ kJmol}^{-1}$

Bond dissociation enthalpy for $\text{Si} - \text{H} = 304 \text{ kJmol}^{-1}$

$\Delta_f H [\text{Si}(g)] = 405 \text{ kJmol}^{-1}$ (roundoff answer to nearest integer)

Question No. 5 (Numerical Type)

The calculated spin-only magnetic moment of X^{n+} is $\sqrt{24}$ B.M. The X^{n+} species can be among the given list of ions. Mark the number of these ions that satisfy the given condition.



Mathematics Single Correct (Maximum Marks: 80)

Question No. 1 (Only One Option Correct Type)

If A is a skew-symmetric matrix of order 2 and B, C are matrices

$$\begin{bmatrix} 1 & 4 \\ 2 & 9 \end{bmatrix}, \begin{bmatrix} 9 & -4 \\ -2 & 1 \end{bmatrix}$$

respectively, then

$$A^3BC + A^5(B^2C^2) + A^7B^3C^3 + \dots + A^{2n+1}B^nC^n$$

- (a) a symmetric matrix
- (b) a skew-symmetric matrix
- (c) an identity matrix
- (d) none of these

Question No. 2 (Only One Option Correct Type)

Let $\vec{x}$ be a vector in the plane containing vectors $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$. If the vector $\vec{x}$ is perpendicular to $(3\hat{i} + 2\hat{j} - \hat{k})$ and its projection on $\vec{a}$ is $\frac{17\sqrt{6}}{2}$, then the value of $|\vec{x}|^2$ is equal to _____.

- (a) 496
- (b) 486
- (c) 476
- (d) 466

Question No. 3 (Only One Option Correct Type)

L_1 and L_2 are two lines whose vector equations are

$$L_1 : \vec{r} = \lambda((\cos \theta + \sqrt{3})\hat{i} + (\sqrt{2}\sin \theta)\hat{j} + (\cos \theta - \sqrt{3})\hat{k})$$

$L_2 : \vec{r} = \mu(a\hat{i} + b\hat{j} + c\hat{k})$, where λ and μ are scalars and α is the acute angle between L_1 and L_2 . If the angle ' α ' is independent of θ then the value of ' α ' is

(a) $\frac{\pi}{6}$

(b) $\frac{\pi}{4}$

(c) $\frac{\pi}{3}$

(d) $\frac{\pi}{2}$

Question No. 4 (Only One Option Correct Type)

If 't' is the period of $f(x)$ satisfying $f(x+5) + f(x) = 0, \forall x \in \mathbb{R}$ and the 4^{th} term in the expansion of $(3 - \frac{2x}{5})^t$ has the greatest numerical value, then $|x|$ belongs to

- (a) $[\frac{45}{16}, \frac{30}{7}]$
- (b) $[\frac{30}{7}, \frac{25}{4}]$
- (c) $[\frac{25}{4}, 9]$
- (d) $[\frac{5}{3}, \frac{45}{16}]$

Question No. 5 (Only One Option Correct Type)

The domain of the function

$$f(x) = \sin^{-1} \left[\log_4 \left(\frac{x}{4} \right) \right] + \sqrt{17x - x^2 - 16} \text{ is....}$$

- (a) $[-1, 1]$
- (b) $[1, 4]$
- (c) $(0, 16]$
- (d) $[1, 16]$

Question No. 6 (Only One Option Correct Type)

Let P_1, P_2 and P_3 be the probabilities of a student passing three independent exams A, B and C respectively. If P_1, P_2 and P_3 are the roots of the equation $20x^3 - 27x^2 + 14x - 2 = 0$, then the probability that the student passes in exactly one of A, B and C is

- (a) $\frac{3}{20}$
- (b) $\frac{7}{20}$
- (c) $\frac{1}{4}$
- (d) $\frac{1}{5}$

Question No. 7 (Only One Option Correct Type)

The smallest positive integer ' n ' such that

$$\frac{1}{\sin 45^\circ \sin 46^\circ} + \frac{1}{\sin 47^\circ \sin 48^\circ} + \cdots + \frac{1}{\sin 133^\circ \sin 134^\circ} = \frac{1}{\sin n^\circ} \text{ is } \text{---}$$

- (a) 2
- (b) 1
- (c) 3
- (d) 4

Question No. 8 (Only One Option Correct Type)

Let $f(x)$ be a differentiable non- decreasing function such that

$$\int_0^x (f(t))^3 dt = \frac{1}{x^2} \left(\int_0^x f(t) dt \right)^3 \forall x \in \mathbb{R} - \{0\} \text{ and } f(1) = 1. \text{ If } \int_0^x f(t) dt = g(x) \text{ then } \frac{xg'(x)}{g(x)} \text{ is.}$$

- (a) Always equal to 1
- (b) Always equal to -2
- (c) May be 1 or -2
- (d) Not independent of x

Question No. 9 (Only One Option Correct Type)

Let $f(x) = x^2 + 4x + a$ and $g(x) = x^2 + 6x + 2a$ be two functions and another function

$h(x) = \frac{f(x)}{g(x)} \forall g(x) \neq 0$. Then find the sum of all integral values of a for which $y = h(x)$ is an onto function.

- (a) 5
- (b) 4
- (c) 6
- (d) 8

Question No. 10 (Only One Option Correct Type)

A bag contains 40 tickets numbered from 1 to 40. Two tickets are drawn from the bag without replacement. The probability that the 2^{nd} ticket is a perfect square given that the 1^{st} ticket was a perfect square is....

- (a) $\frac{1}{6}$
- (b) $\frac{5}{39}$

- (c) $\frac{3}{20}$
- (d) $\frac{1}{8}$

Question No. 11 (Only One Option Correct Type)

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that its derivative f' is continuous and $f(\pi) = -6$. If $F : [0, \pi] \rightarrow \mathbb{R}$ is defined by $F(x) = \int_0^x f(t)dt$, and if $\int_0^\pi (f'(x) + F(x)) \cos x dx = 2$, then the value of $f(0)$ is _____.

- (a) 5
- (b) 3
- (c) 2.5
- (d) 4

Question No. 12 (Only One Option Correct Type)

Let $a_1, a_2, a_3 \dots a_n (n \in \mathbb{N})$ be a sequence such that $a_1 = a_2 = 2$ and $\frac{2a_{n-1}a_n}{a_{n-1}a_{n+1} - a_n^2} = n^3 - n, \forall n = 2, 3, 4, \dots$, find the value of $[S]$, where $S = \sum_{k=2}^{2000} \frac{a_{k+1}}{a_k}$ and $[:]$ denotes G.I.F.)

- (a) 3999
- (b) 2998
- (c) 3998
- (d) 2999

Question No. 13 (Only One Option Correct Type)

Consider $f : \mathbb{R}^+ \rightarrow \mathbb{R}$ such that $f(3) = 1$ for $a \in \mathbb{R}^+$ and

$$f(x) \cdot f(y) + f\left(\frac{3}{x}\right) f\left(\frac{3}{y}\right) = 2f(xy) \forall x, y \in \mathbb{R}^+ \text{ then } f(97) \text{ can be?}$$

- (A) 1
- (B) -1
- (C) 2
- (D) 97

Question No. 14 (Only One Option Correct Type)

Let A and B be two fixed points and P , another point in the plane, moves in such a way that $k_1 PA + k_2 PB = k_3$, where k_1, k_2 and k_3 are real constants. Then which one of the following is not the locus of P .

- (a) a circle, if $k_1 = 0$ and $k_2, k_3 > 0$
- (b) a circle, if $k_1 > 0$ and $k_2 < 0$ and $k_3 = 0$, where $k_1 + k_2 \neq 0$

- (c) an ellipse, if $k_1 = k_2 > 0$ and $k_3 > 0$ and $\frac{k_3}{k_1} > AB$
- (d) a hyperbola, if $k_2 = -1$ and $k_1, k_3 > 0$

Question No. 15 (Only One Option Correct Type)

Find the ordered triplet (A, B, λ) , if

$$\int \frac{2 \cos x - \sin x + \lambda}{\cos x + \sin x - 2} dx = A \log_e(|\cos x + \sin x - 2|) + Bx + C.$$

- (a) $(\frac{1}{2}, \frac{3}{2}, -1)$
- (b) $(\frac{3}{2}, \frac{1}{2}, -1)$
- (c) $(\frac{1}{2}, -1, -\frac{3}{2})$
- (d) $(\frac{3}{2}, -1, -\frac{1}{2})$

Question No. 16 (Only One Option Correct Type)

Find the area bounded by the curve $y = xe^{-x^2}$, the x -axis and the line $x = c$ where $y(c)$ is maximum.

- (a) $\frac{1}{2} \left(1 - \frac{1}{\sqrt{c}}\right)$ sq. units
- (b) $\frac{1}{4} (1 - e^{1/4})$ sq. units
- (c) $\frac{1}{3} (1 - e^{1/3})$ sq. units
- (d) $(1 - e^{1/2})$ sq. units

Question No. 17 (Only One Option Correct Type)

The perpendicular distance from the point $P(3, 5, 2)$ to the line L passing through the point $2i + j$ and parallel to the vector $i + 5j + 2k$ is—

- (a) $\frac{1}{\sqrt{6}}$
- (b) $\frac{2}{\sqrt{6}}$
- (c) $\frac{\sqrt{6}}{\sqrt{5}}$
- (d) $7\sqrt{6}$

Question No. 18 (Only One Option Correct Type)

The locus of the mid-points of the chords of the hyperbola $3x^2 - 2y^2 + 4x - 6y = 0$ which are parallel to the line $y = 2x + 4$ is—

- (a) $3x - 4y = 4$
- (b) $4x - 4y = 3$
- (c) $3y - 4x + 4 = 0$
- (d) $3x - 4y = 2$

Question No. 19 (Only One Option Correct Type)

Let α and β be real and z be a complex number. If $z^2 + \alpha z + \beta = 0$ has two distinct roots on the line $\operatorname{Re}(z) = 1$, then it is necessary that....

- (a) $\beta \in (-1, 0)$
- (b) $|\beta| = 1$
- (c) $\beta \in (1, \infty)$
- (d) $\beta \in (0, 1)$

Question No. 20 (Only One Option Correct Type)

Let $\triangle ABC$ be an isosceles triangle with $AB = AC$. If $B = (0, a)$, $C = (2a, 0)$ and the equation of AB is $3x - 4y + 4a = 0$, then the equation of side AC is

- (a) $y = 8x - 16a$
- (b) $3y = 4x - 8a$
- (c) $x = 2a$
- (d) $y + 8x = 16a$

Mathematics Numerical (Maximum Marks: 20)

Question No. 1 (Numerical Type)

Let the curve C be the mirror image of the parabola $y^2 = 4x$ with respect to the line $x + y + 4 = 0$. If A and B are the points of intersection of C with the line $y = -5$, then the distance between A and B is ____.

Question No. 2 (Numerical Type)

Let $S_r = \{(x, y, z) | x + y + z = 11, x \geq r, y \geq r, z \geq r, x, y, z, r \text{ are integers}\}$ and $n(S_r)$ represents the number of elements in S_r . Then $n(S_2) + n(S_3) + n(S_4) = \underline{\hspace{2cm}}$.

Question No. 3 (Numerical Type)

Consider the differential equation

$$\frac{x + \frac{z^3}{3!} + \frac{z^5}{5!} + \dots}{1 + \frac{z^2}{2!} + \frac{z^4}{4!} + \dots} = \frac{dx - dy}{dx + dy}.$$

If $y(0) = 1$ and the solution of the differential equation is of the form

$$2yf(x) = me^{2x} + n,$$

evaluate $(m + n)f(0)$.

Question No. 4 (Numerical Type)

The difference between any two consecutive interior angles of a polygon is 5° . If the smallest angle is 120° , then the number of the sides of the polygon is _____.

Question No. 5 (Numerical Type)

$$\lim_{x \rightarrow 1} \frac{\sin^2(x^3 + x^2 + x - 3)}{1 - \cos(x^2 - 4x + 3)}$$

has the value equal to _____.

(ANSWER KEY)

Physics

Q1 []	Q2 []	Q3 []	Q4 []	Q5 []
Q6 []	Q7 []	Q8 []	Q9 []	Q10 []
Q11[]	Q12 []	Q13 []	Q14 []	Q15 []
Q16[]	Q17 []	Q18 []	Q19 []	Q20 []

Q21 []	Q22 []	Q23 []	Q24 []	Q25 []
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Chemistry

Q1 []	Q2 []	Q3 []	Q4 []	Q5 []
Q6 []	Q7 []	Q8 []	Q9 []	Q10 []
Q11[]	Q12 []	Q13 []	Q14 []	Q15 []
Q16[]	Q17 []	Q18 []	Q19 []	Q20 []

Q21 []	Q22 []	Q23 []	Q24 []	Q25 []
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Mathematics

Q1 []	Q2 []	Q3 []	Q4 []	Q5 []
Q6 []	Q7 []	Q8 []	Q9 []	Q10 []
Q11[]	Q12 []	Q13 []	Q14 []	Q15 []
Q16[]	Q17 []	Q18 []	Q19 []	Q20 []

Q21 []	Q22 []	Q23 []	Q24 []	Q25 []
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